

Dynamic behavior of metal droplet impact on dry smooth wall: SPH simulation and splash criteria

Tianyu Ma, Dawei Chen^{*}, Haiquan Sun, Dongjun Ma, Aiguo Xu, Pei Wang

Institute of Applied Physics and Computational Mathematics, Key Laboratory of Computational Physics, Beijing, 100094, China

ARTICLE INFO

Article history:

Received 25 August 2020
Received in revised form 4 November 2020
Accepted 30 January 2021
Available online 3 February 2021

Keywords:

SPH
Metal droplet
Dry smooth wall
Physical regulation
Splash criteria

ABSTRACT

Droplet impingement is a widespread phenomenon in nature and industrial production. Different from the problem of light drop/wall impact, which has been investigated early and sufficiently, few researches are found on the dynamic behavior and modeling of heavy drop impingement, which is widely existed in the field of plasma spray, 3D printing and ejecta mixing caused by detonation, etc. In this paper, the behavior of different metal drops impinging on dry smooth wall is studied based on Smoothed Particle Hydrodynamics (SPH) method. A newly splash threshold is proposed for heavy drop impact, and the difference of splash criteria between light drop and heavy drop is analyzed. The results show that incident kinetic energy is an important factor to determine the outcomes of heavy droplet impact. When the incident kinetic energy is relatively low, the metal droplet would spread, and the maximum spreading radius increases with the increase of kinetic energy. With higher incident kinetic energy, the droplet would splash, and the proportion of secondary droplets increases with the increase of incident kinetic energy. Higher droplet viscosity will hinder its spreading, but when splash occurs, it will promote the increase of secondary droplet. The increase of surface tension would suppress droplet spreading rate, but it has little effect on the maximum spreading radius. The splash criteria of heavy droplet impinging on dry smooth wall is proposed as $K = Oh \cdot Re^{1.5346} = 3341.95$. The splash threshold of heavy droplet is higher than that of light droplet within the range of $Re < 4000$, owing to the higher surface tension of heavy droplet. When $Re > 4000$, the outcomes of light droplet impact show strong randomness, thus no uniform splash threshold can be found, while the metal droplet has a uniform splash threshold within the range of $Re < 100000$.

© 2021 Elsevier Masson SAS. All rights reserved.

1. Introduction

Droplet impingement is commonly found in energy and power engineering, aviation industry and daily life [1]. It is also a key problem of spray combustion in internal combustion engine, thermal spraying, agriculture irrigation, as well as micro ejection of metal droplet during implosion process. Being an intercross topic, droplet impaction involves multiphase flow, heat and mass transfer and interface dynamics. When a droplet hits the wall, the incident and boundary conditions are responsible for the droplet deformation, resulting in various outcomes [2,3].

The investigation of light droplet impact on wall was began as early as last century [4]. Wachters and Weaterling [5] found that the water droplet did not breakup when $We < 30$, and it began to rebound and breakup when $30 < We < 80$. For $We > 80$, the droplet would splash. An early droplet impingement model was proposed by Naber and Reitz [6] based on the experimental result from Ref. [5]. The incident droplet was treated as wall jet,

which moves along the wall without breakup. By comparing the three impact modes including stick, rebound and wall jet, the wall jet model could provide reasonable simulation results. Mundo et al. [7,8] take $K = Oh \cdot Re^{1.25} = 57.7$ as the splash criteria based on Schmidt et al. [9]'s experimental results. This splash threshold is widely used on the judgment of droplet deposition/splash. Senda et al. [10,11] considered the effect of wall temperature on fuel spray impingement, and indicated that film formation should be taken into account when the wall temperature is lower than the fuel boiling point. When the wall temperature reaches the boiling point, heat transfer and breakup became the main physical process. $We = 300$ was chosen as the splash criteria in their model. O'Rourke and Amsden [12] improved the splash criteria proposed by Mundo et al. [7] and Eq. (1) was obtained as a splash criteria.

$$K = \sqrt{\frac{\rho u_0^2 d_0}{\sigma} \frac{1}{\min\left(\frac{h_0}{d_0}, 1\right) + \frac{\delta_{bl}}{d_0}}} > 57.7 \quad (1)$$

^{*} Corresponding author.

E-mail address: chen_dawei@iapcm.ac.cn (D. Chen).

Nomenclature**Abbreviations**

SPH	Smoothed Particle Hydrodynamics
YSZ	Yttria Stabilization Zirconia
CSF	Continuous Surface Force

Roman symbols

c_s	Numerical speed of sound
d_0	Initial droplet diameter
E_k	Droplet kinetic energy
$f(\mathbf{r})$	Physical quantity in SPH
$(f_s)_i$	Surface tension of particle i
$f_{ij}^n R_{ij}$	Artificial force
FW_{ij}	Wall force
g_i	Gravitational acceleration of particle i
h	Smoothing length
h_0	Film thickness
K	Droplet splash criteria
L	Wall length
L_d	Droplet spreading diameter
m_i	Mass of particle i
$\mathbf{n}_i$	Surface normal direction of particle i
$\mathbf{n}_w$	Wall normal direction
N	Particle number
Oh	Ohnesorge number
p	Pressure
p_i	pressure of particle i
q	Non-dimensional particle distance
$\mathbf{r}$	Space vector
Re	Reynolds number
t	Time
T_L	Droplet temperature
T_{Leid}	Leidenfrost temperature
T_m	Droplet melting point temperature
T_w	Wall temperature
u_0	Droplet incident velocity
u_i	Velocity of particle i
u_{ij}	Relative velocity between particle i and particle j
u_{max}	Maximum particle velocity
W	Kernel function
We	Weber number
x_i	Position of particle i

Greek symbols

γ	Exponent of state equation
δ_{bl}	Boundary layer thickness
Δx	Initial particle spacing
μ_i	Dynamic viscosity of particle i
ρ_0	Reference density
ρ_i	Density of particle i
σ	Coefficient of surface tension
τ	Non-dimensional time ($=t \times u_0/d_0$)
Φ	Spread factor ($=L_d/d_0$)
Ω	Particle supporting field

When the film thickness $h_0 = 0$, and take the boundary layer as $\delta_{bl} = d_0/\sqrt{Re}$, the splash criteria of Eq. (1) could be written as $K = We^{0.5} Re^{0.25}$. Vander War et al. [13] and Palacios et al. [14] summarized the droplet splashing criteria of different light materials, including water, ethanol, glycerin and alkanes. Similar threshold of splash was found although the physical properties were different. Staat et al. [15] studied the regimes of ethanol droplet impingement. When the wall temperature is less than the Leidenfrost Temperature (T_{Leid}), the droplet outcome changes from spread to splash with the increase of We number. When the wall temperature is higher than T_{Leid} , ethanol drop would transform from rebound to splash with the increase of We number. By summarizing the outcome regimes of different light droplets impinging on low temperature wall, it can be found that the splash criteria can be expressed as $OhRe = 17$ [16] when the incident $Re < 4000$. While with wall temperature higher than the boiling point, the droplet outcomes are affected by the near wall heat transfer, and the droplet splash can be judged by the incident We and the wall temperature.

In recent years, the problem of single light droplet hitting a dry smooth wall has been well investigated, and a relatively clear splash threshold can be obtained. Besides, heavy drop impingement such as metal drop has arouse more and more interest owing to its wide application in industry. With different physical properties, the morphology evolution of heavy drop after impact would be different from light drop. However, few experimental and numerical investigations are found on heavy droplet impingement.

Aziz et al. [17] studies the impinging process of a 2.7 mm tin droplet with the velocity of 1~4 m/s. Prompt splash was observed at the front edge with the increase of incident velocity because of Rayleigh–Taylor instability. Tabbara et al. [18] studied the outcome of tin droplet impact with different velocities. The time scale of droplet spreading was found to be smaller than that of the heat transfer when the incident velocity is higher than 100 m/s, thus the solidification can be ignored. Mehdizadeh et al. [19] and Dhiman et al. [20,21] investigated the influence of wall temperature on tin drop impingement. With wall temperature far below the melting point, the front edge of drop could freeze which induces splash. By raising the wall temperature, the splash is suppressed. Wei et al. [22] and Chen et al. [23] studied the impingement of Yttria Stabilization Zirconia (YSZ) drop. It was found that surface tension play a dominant role for droplet splash when $Oh < 0.2$, and the splash threshold is $We \sim 1.35 \times 10^4 \pm 100$. While droplet viscosity would lead the splash when $Oh > 0.2$, and the splash threshold is $Re \sim 450 \pm 20$. Different from Wei et al. [22] and Chen et al. [23] Wang et al. [24] took $K = We^{0.5} Re^{0.25}$ as the splash criteria for YSZ drop impingement, which is the same as the criteria of light drop impingement.

For the simulation of droplet impingement, Smoothed Particle Hydrodynamic (SPH) method has been widely used since 1990s. Being a meshless method, it is able to provide reasonable results quickly compared to other interface capturing methods such as VOF. Xu et al. [25] successfully simulated the successive impinging of two 2D water droplet impact and the impact of one 3D droplet. The outcomes and the variation of pressure field can be well estimated with incompressible SPH method. Yang and Kong [26] studied the process of a light weight droplet impacting on a dry solid surface with 3D SPH by coupling a new surface tension model and artificial viscosity. It is found that the droplet spreads easier with higher We number. The impact of ellipsoidal droplet is also studied, indication that an oblate drop can cause splash easily. Abdollahi et al. [27] simulated the single droplet dynamic when hitting a thin film with a weakly compressible multiphase SPH. The crown was captured although the rim part was slightly breakup. Dong et al. [28] and Hao

et al. [29] simulated the impact process of a droplet on elastic beams. Typical regimes, such as spherical-shaped rebound, pancake-shaped rebound and splashing of water droplet were identified. The relationship between impact force and droplet shape was discussed by Zhang et al. [30] with SPH method. The peak impact force increases with the increase of “horizontal-to-vertical ratio”. However, no researches on metal droplet impact can be found with SPH method.

According to the review above, it is concluded that the detailed regulation of metal droplet impinging on dry smooth wall is not fully understood. And the splash criteria for light droplet impingement was still adopted for heavy droplet in some researches [24,31]. However, deeper understanding of the dynamic behavior after heavy droplet impingement is needed in the field of plasma spray, 3D printing and ejecta mixing caused by detonation [32], etc.

In this paper, metal droplet/wall dynamic behavior is studied based on SPH simulation. The morphological evolution after drop impact are discussed and analyzed. The model of heavy droplet splash threshold is built by combining the related experimental results and numerical results. And the similarities and differences of splash criteria between light drop and heavy drop are discussed.

2. Numerical calculation method and validation

2.1. SPH method

Owing to the phenomenon of large deformation and breakup during droplet impingement, particle approach is more suitable. In SPH method, the fluid is discretized as a group of particles. The motion of fluid is represented by the movement and distribution of particles. Compared with other interface capturing method, SPH is much more efficient on analyzing the droplet breakup process. In addition, the details of droplet outcome after impact can be captured by tracing the movement of particles.

In this paper, SPH method is used to study the problem of metal droplet impinging on dry smooth wall. By discretizing the droplet as a group of particles, the physical parameters can be calculated by a field integral formula as shown in Eq. (2).

$$f(\mathbf{r}) \approx \int_{\Omega} f(\mathbf{r}') W(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' \quad (2)$$

in which Ω is the supporting area where particles interact with each other, $W(\mathbf{r} - \mathbf{r}', h)$ the smoothing kernel function, $\mathbf{r}$ represents the position of particle and h defines the smoothing length of supporting area. The interaction between particles is mainly controlled by the smoothing kernel function. In this paper, the cubic spline kernel function is used for droplet simulation as shown in Eq. (3).

$$W(q, h) = \alpha_d \times \begin{cases} \frac{2}{3} - q^2 + \frac{1}{2}q^3, & 0 \leq q < 1 \\ \frac{1}{6}(2 - q)^2, & 1 \leq q < 2 \\ 0, & 2 \leq q \end{cases} \quad (3)$$

where $\alpha_d = \frac{15}{7\pi h^2}$ in two dimensional calculation, and q represents the distance between particles nondimensionalized by the smoothing length h .

Based on SPH approximation method described above, the continuity equation, momentum equation and the particle position updating equation of single metal droplet impingement are

discretized and shown in Eq. (4).

$$\begin{cases} \frac{d\rho_i}{dt} = \sum_{j=1}^N m_j u_{ij} \cdot \frac{\partial W_{ij}}{\partial \mathbf{x}_j} \\ \frac{d\mathbf{u}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2} + f_{ij}^n R_{ij} \right) \frac{\partial W_{ij}}{\partial \mathbf{x}_i} \\ \quad + \sum_{j=1}^N m_j \frac{\mu_i + \mu_j}{\rho_i \rho_j} u_{ij} \frac{1}{r_{ij}} \frac{\partial W_{ij}}{\partial r_{ij}} - (f_s)_i - g_i \\ \frac{d\mathbf{x}_i}{dt} = \mathbf{u}_i \end{cases} \quad (4)$$

in which $\mathbf{x}_i$, $\mathbf{u}_i$, p_i , $(f_s)_i$ and g_i are the position, velocity, pressure, surface tension and gravitational acceleration of particle i , respectively. In order to eliminate the tensile instability of SPH, the artificial stress $(f_{ij}^n R_{ij})$ proposed by Monaghan [33] and Gray [34] is used in Eq. (4), where $n = 4$ for this two-dimensional system.

In order to solve the particle pressure in momentum equation, an artificial state equation is applied as shown in Eq. (5),

$$p = \frac{c_s^2 \rho_0}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right) \quad (5)$$

where γ is the exponential relationship between pressure and density, and here $\gamma = 7$. ρ_0 is the reference density of particle. c_s is numerical sound speed, which is set as 10 times of the maximum droplet velocity.

The continuous surface force (CSF) model proposed by Brackbill et al. [35] is applied to calculate droplet surface tension. The surface normal direction and surface curvature, which change continuously, are solved by the newly developed color function [36]. So that the final surface tension per unit mass can be expressed in Eq. (6)

$$(f_s)_i = - \frac{\sigma}{\rho_i} (\nabla \cdot \mathbf{n})_i n_i \quad (6)$$

where σ is the coefficient of surface tension, and $\mathbf{n}_i = (n_{xi}, n_{yi}, n_{zi})$ is the surface normal vector. And the surface normal vector and surface curve can be calculated as:

$$\begin{aligned} n_{\alpha i} &= \left[\sum_{j=1}^n (\bar{c}_j - \bar{c}_i) W_{ij,\beta} \frac{m_j}{\rho_j} \right] \left[\sum_{j=1}^n (x_j^\alpha - x_i^\alpha) W_{ij,\beta} \frac{m_j}{\rho_j} \right]^{-1} \\ \hat{n}_{\alpha,\gamma i} &= \left[\sum_{j=1}^n (\hat{n}_{\alpha j} - \hat{n}_{\alpha i}) \cdot W_{ij,\beta} \frac{m_j}{\rho_j} \right] \left[\sum_{j=1}^n (x_j^\gamma - x_i^\gamma) W_{ij,\beta} \frac{m_j}{\rho_j} \right]^{-1} \end{aligned} \quad (7)$$

where $\bar{c}_i = \sum_{j=1}^n \frac{m_j}{\rho_j} C_j W_{ij}$, and the subscript α, β and γ represent the spatial orientation, while i and j are the particles mark.

For the numerical simulation of droplet impingement, the most important model is the boundary force model which influences the accuracy of spread/splash outcome. In order to prevent particles from penetrating the solid wall, the interaction between drop and wall is simulated by the combination of virtual particles and penalty function. And the penalty function is referenced from the latest algorithm [37] as shown in Eq. (8).

$$FW_{ij} = \begin{cases} - \left[u_{\max}^2 \cdot \frac{\min(\mathbf{u}_{ij} \cdot \mathbf{n}_{Wj}, -1) W_{ij} H_{ij} \mathbf{n}_{Wj}}{|\mathbf{x}_{ij} \cdot \mathbf{n}_{Wj}|} \right] & \mathbf{u}_{ij} \cdot \mathbf{n}_{Wj} < 0 \\ 0 & \mathbf{u}_{ij} \cdot \mathbf{n}_{Wj} \geq 0 \end{cases} \quad (8)$$

where $H_{ij} = [(h_i + h_j)/2]^2$, and $\mathbf{n}_W$ represents the wall normal vector. In the computation, leap-frog format [38] is used for

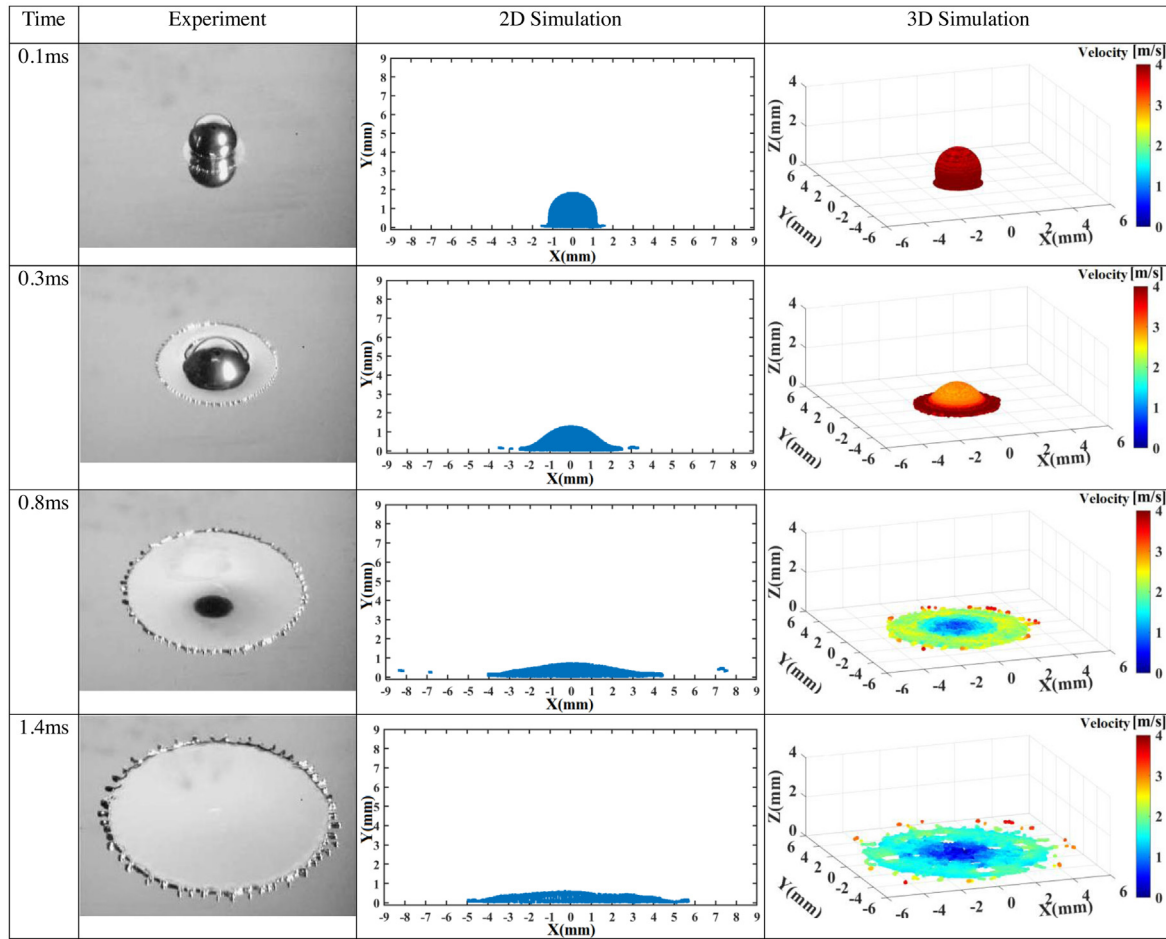


Fig. 1. Impact of 2.2 mm Sn droplet on a flat surface with 4 m/s.

the time integral, and the time step is estimated as the minimum value of the time step expression based on external force, Brackbill's expression [35] based on surface tension and Morris' expression [39] based on viscosity diffusion:

$$\begin{aligned} \Delta t &\leq \min \left(0.25 \frac{h_i}{F_i} \right) \\ \Delta t &\leq \min \left[0.25 \left(\frac{\rho h_i^3}{2\pi\sigma} \right)^{0.5} \right] \\ \Delta t &\leq \min \left(0.125 \frac{\rho h_i^2}{\mu} \right) \end{aligned} \quad (9)$$

2.2. Computing validation

The SPH method has been used for the quantitative analysis of light drop impingement and the effectiveness of such algorithm was proved to be reliable [25–30]. In this simulation, the SPH method is used to simulate the impinging process of heavy droplet, where there are few quantitative experimental results. So, typical experimental results of a Sn droplet impingement case from [40] are used for the validation. 9600 virtual particles are arranged in three layers to simulate the solid wall. And the drop is simulated by 5016 particles which are uniformly distributed. As every particle can be tracked with SPH method, there is no need to set calculation boundary. The computational setup is listed in Table 1.

Both 2D simulation and 3D simulation are conducted based on this SPH code and compared with experimental results for cross

verification. Fig. 1 shows the comparison between experiments and simulation of a 2.2 mm Sn droplet impinging on a flat wall with the velocity of 4 m/s. A thin film is observed which gradually formed after the impact. With the spreading of film, finger-like structure and a little splash is found near the front edge. Owing to the impact force in the central part of droplet, the rim velocity is higher than the middle part velocity. Both 2D and 3D SPH simulation are able to capture the spreading process and the deformation of Sn drop observed in the experiment. Besides, the boundary jets at the rim of spreading could also be captured by both simulations, but the boundary splash outcomes of 3D results are more obvious compared to the experimental results as shown in Fig. 1.

In order to further validate the accuracy of SPH method, the evolution process of droplet spread factor is compared with the experimental results quantitatively. By comparing the development of spread factor after droplet impact between experiment and simulation, it can be seen that both the film spreading rate and the maximum spreading diameter are reasonably captured by 2D and 3D SPH simulation accurately, as shown in Fig. 2. Thus, the impact of metal droplet impingement can be simulated effectively and correctly with SPH.

3. Simulation analysis of metal droplet impingement

In order to explore the influence of incident kinetic energy and droplet material on the outcome of metal drop impingement, 30 μm drops of different materials including Sn, Pb, Ga and Al

Table 1
Initial parameters of droplet/wall impingement.

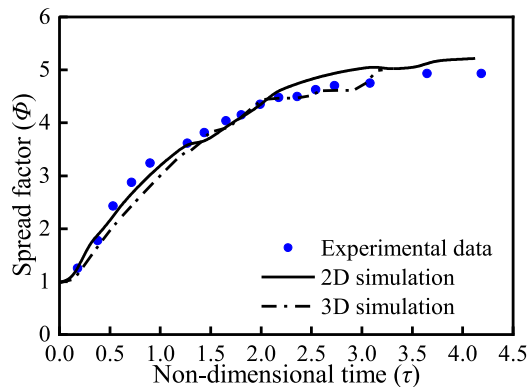
Parameters	Value
Diameter d_0 (mm)	2.2
Velocity u_0 (m/s)	4
Density ρ (kg/m ³)	6970
Viscosity μ (Pa s)	1.92×10^{-3}
Surface tension σ (N/m)	0.526
Smoothing length h	$1.3\Delta x$
Initial particle spacing Δx (m)	3.67×10^{-5}
Wall length L (mm)	11

Table 2
Physical properties of droplet materials [41].

Material	ρ (kg/m ³)	σ (N/m)	μ (Pa s)
Pb	10 670	0.458	2.61×10^{-3}
Ga	6100	0.718	1.94×10^{-3}
Al	2570	1.07	2.10×10^{-3}

Table 3
Case setup in droplet simulation.

Material	d_0 (μ m)	u_0 (m/s)	Re	We	Oh	E_k (μ J)
Sn	30	10	1090.9	39.8	0.00578	0.00493
Sn	30	30	3272.7	357.8	0.00578	0.04434
Sn	30	100	10 909.1	3975.3	0.00578	0.49268
Sn	30	300	32 727.3	35 777.6	0.00578	4.43412
Pb	30	8.08	993.5	45.9	0.00682	0.00492
Pb	30	80.82	9912.5	4565.2	0.00682	0.49265
Ga	30	10.69	1008.5	29.1	0.00535	0.00493
Ga	30	106.89	10084	2912.1	0.00535	0.49265
Al	30	16.47	1149.1	19.5	0.00385	0.00493
Al	30	164.68	11 489.3	1954.1	0.00385	0.49266

**Fig. 2.** Evolution of spread factor during the impingement of 2.2 mm Sn droplet with 4 m/s.

are selected for a series of simulation. The physical properties of Sn drop are shown in Table 1, while the properties of the other metals are shown in Table 2. The effect of kinetic energy is investigated by changing the droplet incident velocity. And the influence of typical physical properties such as surface tension and viscosity are explored by comparing the impact behavior of different kinds of droplets with the same incident kinetic energy. For example, with similar viscosity and different surface tension, Al droplet and Sn droplet are compared to study the influence of droplet surface tension on their impact process. Besides, Ga droplet is compared with Sn droplet as they own similar physical properties. The details of case setup are listed in Table 3.

Fig. 3 shows the development of a 30 μ m Sn droplet impinging on dry smooth wall with different incident velocity. No splash is found when the velocities are 10 m/s and 30 m/s, which can be seen in Fig. 3(a) and (b). The velocity of the central part of

droplet decreases rapidly after impact because of the wall force. But the outside of droplet spreads quickly along the wall. With the increase of incident kinetic energy, the velocity of the droplet outside part is also increased, resulting in larger spreading radius. When the drop impacts with the speed of 100 m/s and 300 m/s, splash occur as shown in Fig. 3(c) and (d). Due to the high incident energy, small jets are formed with a certain angle against the wall during the early stage of impingement. The jets break up later which are transformed into the secondary droplets. With the increase of incident velocity, the velocity of droplet outside part and secondary droplets are also increased.

Fig. 4 shows the influence of incident kinetic energy on the spreading process by comparing the spread factor of Sn droplets with different incident kinetic energy. In the case of droplet spreading, both the film expansion rate and the maximum spreading radius increase with the rise of impact velocity. This is because higher incident energy creates faster outside jet flow, which could overcome the droplet viscosity and surface tension, resulting in larger spreading radius.

Fig. 5 shows the influence of incident kinetic energy on the mass fraction and energy fraction of secondary droplets. The mass of splashing secondary drop are obtained by summing up the edge particles flying away from wall (defined as the particles who's high is larger than 1/10 droplet diameter and with normal velocity against wall). And the kinetic energy of secondary drop is calculated as the total kinetic energy of flying particles divided by the droplet incident kinetic energy. One can well perceive that with the increase of impact kinetic energy, the splashing mass and energy fraction are similar during early stage ($\tau < 1.5$), but they are increased at last. That is to say, if the incident velocity is higher, there would be more kinetic energy separating the secondary droplet from the main droplet.

Fig. 6 shows the impact process of 30 μ m Ga, Pb and Al droplet with the same low incident kinetic energy. During the droplet spreading, the moving film need to overcome both the

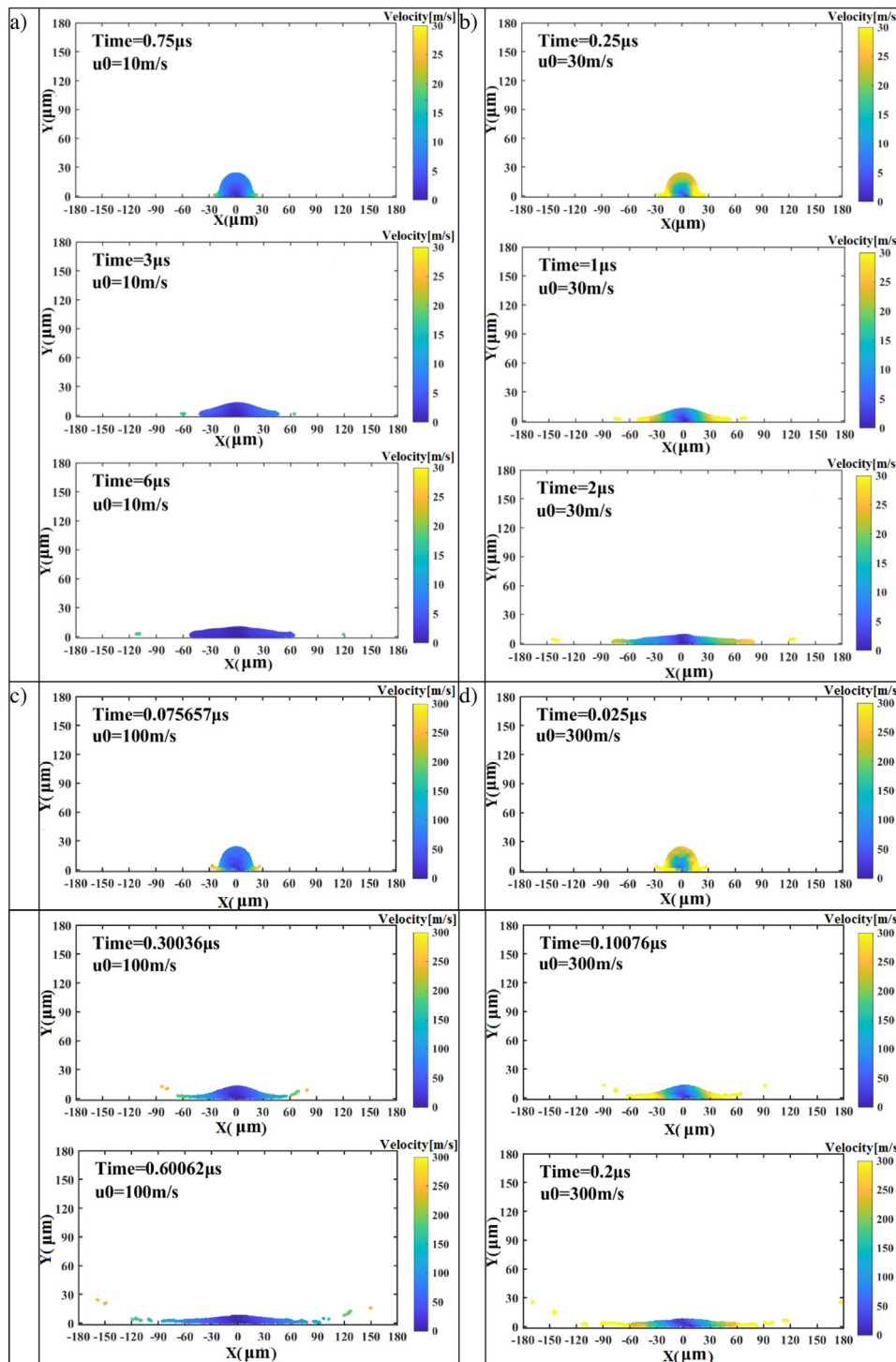


Fig. 3. Development of a 30 μm Sn droplet impact with different velocity ($\tau = 0.25, 1$ and 2).

droplet viscosity and the surface tension. With similar viscosity and surface tension, the spreading process of Ga drop has much in common with Sn drop, as shown in Figs. 6(a) and 3(a). Comparing the impact process of Pb and Ga drop, it is observed that the outside spreading speed of Pb droplet is lower than that of Ga owing to higher viscosity, so it spreads relatively slow as shown in Fig. 6(a) and (b). In addition, the comparison between Al drop and Ga drop shows that with higher surface tension the spreading process of Al drop is relatively delayed, as can be seen in Fig. 6(c) and (a).

Fig. 7 shows the development of spread factor of droplets with different materials and the same incident kinetic energy. As expected, the spreading process of Sn drop and Ga drop are almost the same. Comparing Sn droplet and Pb droplet, it is found that both the spreading rate and the maximum spread factor are reduced with the increase of viscosity. This is because higher droplet viscosity consumes more kinetic energy, which limits the droplet spreading speed and the expansion area. This regulation is also agree with what Qin et al. [46] and Liang et al. [47] found in their early research indicating that the maximum spread factor decrease with the increase of droplet viscosity. Comparing the

Table 4
Summary of investigation on heavy droplet/wall impingement.

Reference	Material	d_0 (mm)	u_0 (m/s)	E_k (μ J)	Re	We
Shakeri et al. [40]	Sn	2.2	4	311	3.2E4	467
Aziz et al. [17]	Sn	2.7	1~4	36~575	9818~3.93E4	36~573
Mostaghimi et al. [42]	Ni/Al	0.06/0.025	73/105	2.38/0.116	7892/267.5	1419/1198
Ghafouri et al. [43]	Sn	2.2	2.5	121	2E4	182
Mehdizadeh et al. [19]	Sn	0.6	10~40	39.4~630	2.18E4~8.73E4	795~12721
Dhiman et al. [20]	Sn	0.6	10~30	39.4~354	2.18E4~6.54E4	795~7156
Dhiman et al. [21]	Sn	0.55	10~40	30.3~485.7	2E4~8E4	729~11661
Xue et al. [44]	Al/Ni	4/0.05	3/72	388/1.34	27907/ 6000	86/1152
Tabbara et al. [18]	Sn	2.2	4~400	311~3.1E6	2.79E4~2.79E6	434~4.34E6
Wei et al. [22]	YSZ	0.004~0.023	226.5~584.4	2.0E~2~3.56	231~1459	11095~8.75E4
Chen et al. [23]	YSZ	0.005~0.179	226.5~584.4	5.71E~2~0.412	414~600.3	1.13E4~3.05E4
Shen et al. [45]	YSZ	0.012	518	0.715	1317	4.41E4
Wang et al. [24]	YSZ	0.025~0.03	465~518	5.2~11.2	1863~2773.2	8.31E4~1.12E5

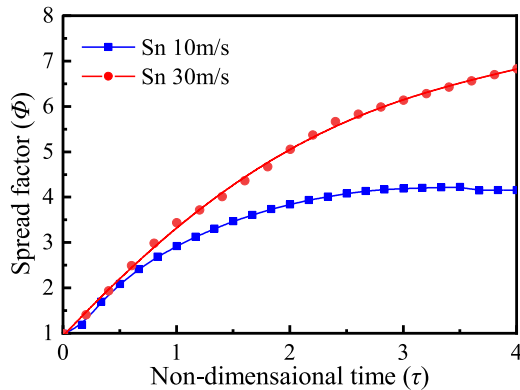


Fig. 4. Influence of kinetic energy on Sn droplet spread factor.

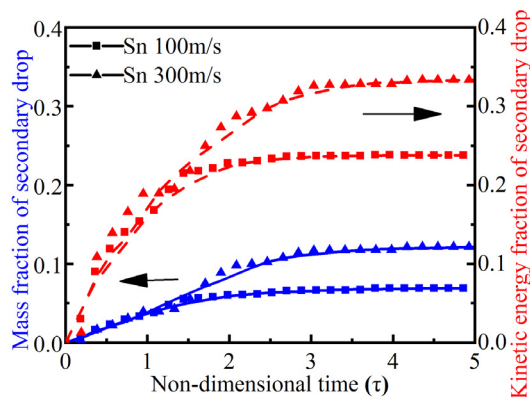


Fig. 5. Influence of incident kinetic energy on the mass and energy fraction of secondary droplets (The shapes represent the different incident velocity as shown in the legend; the red lines are the kinetic energy fraction of secondary drop and the blue lines are the mass fraction of secondary drop).

spread process between Sn droplet and Al droplet, it can be seen that although the maximum spread factor is not affected by the surface tension, the spreading rate is restrained by the increased surface tension. As the surface tension tends to prevent the expansion of surface area during the droplet spread, Al drop needs to overcome more surface energy owing to its higher surface tension. According to the analyze above, droplet viscosity is the main factor influencing the maximum spreading radius.

Fig. 8 shows the impact process of 30 μ m Ga, Pb and Al droplet with the same high incident kinetic energy. Similarly, the outcomes of Ga droplet and Sn droplet are resemblable after impingement as shown in Figs. 8(a) and 3(c). Comparing the splashing process between Pb droplet and Ga droplet as shown in

Fig. 8(b) and (a), it is observed that higher viscosity induces more secondary droplets. The contrast of Al droplet and Ga droplet indicates that the surface tension has no significant effect on the secondary drops at early stage ($\tau < 2$), as shown in Fig. 8(c) and (a).

Fig. 9 displays the development of mass and kinetic energy ratio of secondary droplets of cases shown in Fig. 8. With similar viscosity and surface tension, the mass and kinetic fraction of Sn droplet and Ga droplet are approximately the same. Comparing Pb droplet and Sn droplet, which have similar surface tension but different viscosity, it is found that with the increase of viscosity, both the splashed mass ratio and the kinetic energy ratio of secondary drop are increased apparently. Although owing to the highest surface tension, the viscosity of Al droplet goes between Pb and Ga. And the result shows that the final splashed mass and kinetic energy ratio of Al droplet are also between these two droplets. Based on the description above, it is demonstrated that higher droplet viscosity would promote splash.

4. Splash criteria of heavy drop impingement

In order to make the splash threshold of heavy droplet impact clear, the outcomes of various heavy droplet impingement are investigated based on the numerical research above and available experimental results. The experimental results are collected from several literatures where the size of the incident droplet vary from micron scale to millimeter scale, as shown in Table 4.

Fig. 10(a) shows the heavy droplet behavior after impingement and the splash criteria by summering the numerical and experimental results, in which the red points represent splash while the blue ones represent non-splash. In the Re - Oh logarithmic distribution diagram of droplet state, an obvious boundary of splash and non-splash can be seen in Fig. 10(a). This indicates that the heavy droplet would splash when its exponential product of Oh and Re reaches a certain value. So that the splash criteria can be expressed by the combination of Oh number and Re number. By fitting the boundary line, the splash criteria is obtained in Eq. (10).

$$K = Oh \cdot Re^{1.5346} \quad (10)$$

where $K = 3341.95$. Fig. 10(b) compares the splash criteria between heavy droplet and light droplet, where the threshold of light droplet is relatively clear when $Re < 4000$ according to Ref. [16]. It can be seen that the splash criteria of both heavy droplet and light droplet can be expressed by the combination of Oh and Re , though their physical properties are different. However, the splash threshold of heavy droplet is higher than that of light droplet within the range of $Re < 4000$. Namely that more incident energy is necessary for a heavy droplet to splash under the same Oh number condition. This is because heavy droplet usually has higher surface tension coefficient, making it much

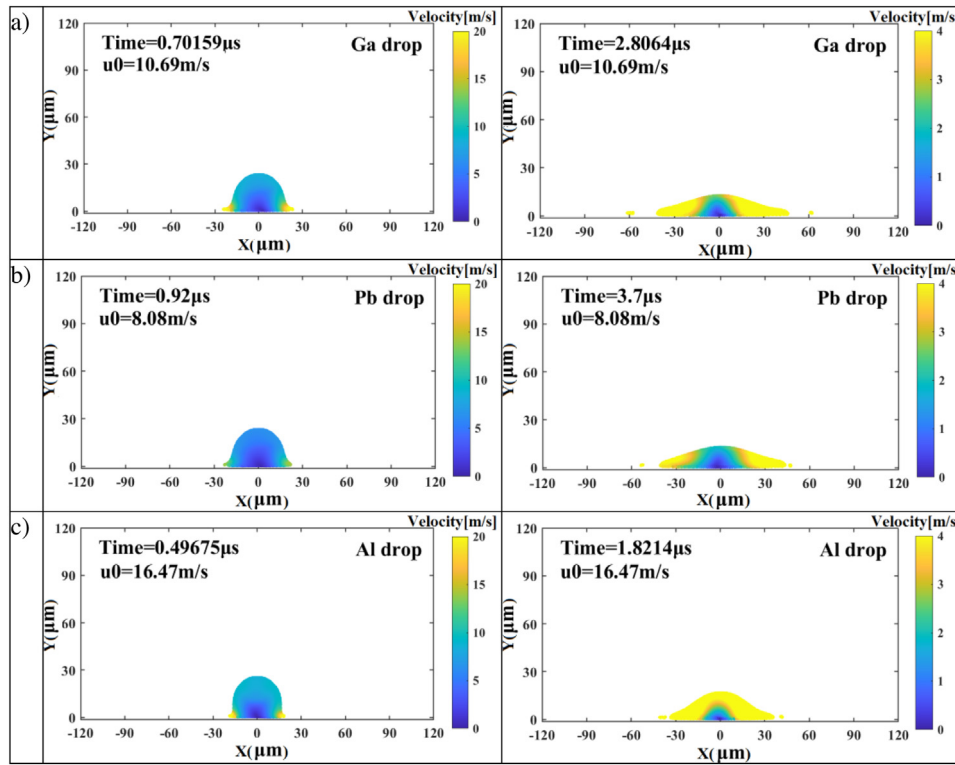


Fig. 6. Impact of 30 μm Ga, Pb and Al drop with low incident energy ($\tau = 0.25$ and 1).

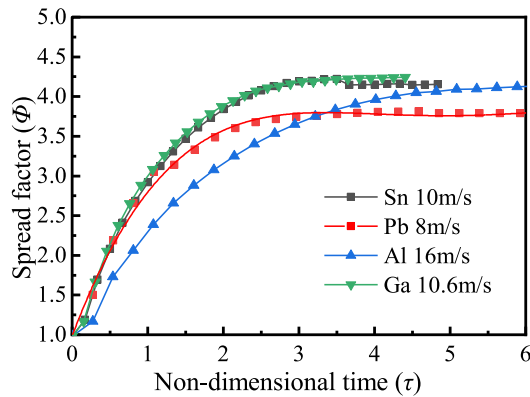


Fig. 7. Influence of droplet material on spread factor under the same incident kinetic energy.

harder to splash. In addition, when the incident Re is higher than 4000, the splash threshold of light droplet is no longer clear [16], and the thresholds proposed by early researcher are quite different from each other. But for heavy droplet, the splash threshold is relatively stable within range of $Re < 100\,000$.

To further verify the effectiveness of heavy droplet splash criteria on the judgment of droplet outcomes, the impingement of Sn droplets with different sizes are simulated based on SPH method. Different cases are setup with varied K values as shown in Table 5. The K values of case 1, 2 and 4 are less than 3341.95, and are predicted to be spread based on the model above. While the K values of case 3 and 5 are higher than 3341.95, which are estimated to be splash. According to the simulation results, the outcomes of Sn drop at $\tau = 2$ is shown in Fig. 11. It can be seen that the simulated results agree with the droplet outcomes estimated by the splash criteria model. Besides, it is indicated

that smaller drop requires higher incident energy to splash by comparing the results of 0.5 mm and 2 μm drop impingement.

The outcome regime and splash criteria of heavy droplet are further discussed with different wall temperature. Fig. 12 shows droplet behavior and splash threshold of heavy droplet impact on dry wall with different temperature. Firstly, it is claimed that the splash criteria of the light drop impingement: $K = We^{0.5} \cdot Re^{0.25}$ no longer fits the splash judgment of heavy droplet impact, as shown in Fig. 12(a). If a constant value is used as the threshold, it would overestimate the splash boundary at higher wall temperature or under estimate the splash boundary at lower wall temperature. However, the proposed splash criteria ($K = Oh \cdot Re^{1.5346}$) can distinguish the outcomes of heavy droplet impingement at different wall temperature. For heavy droplet impinging on dry smooth wall cases, the drop would splash when K reaches 3341.95, otherwise it would not splash. Secondly, it is found that for the drop impingement with the same material, the splash states are gradually suppressed with the increase of wall temperature [19,24].

5. Conclusions

In this paper, the problem of single metal droplet impinging on a dry smooth wall is studied based on SPH method. The influence of incident kinetic energy and droplet material on the outcome are also analyzed. And a splash threshold is given for the judgment of heavy droplet splash. Several conclusions are listed as follows.

1. The impact process of metal droplet is analyzed in detail based on SPH method. The incident kinetic energy is found to be the key factor which influences the droplet outcomes. When the kinetic energy is relatively low, the droplet spreads along the wall, and higher incident energy results in larger maximum spreading radius. Under the same incident energy condition, the maximum spread

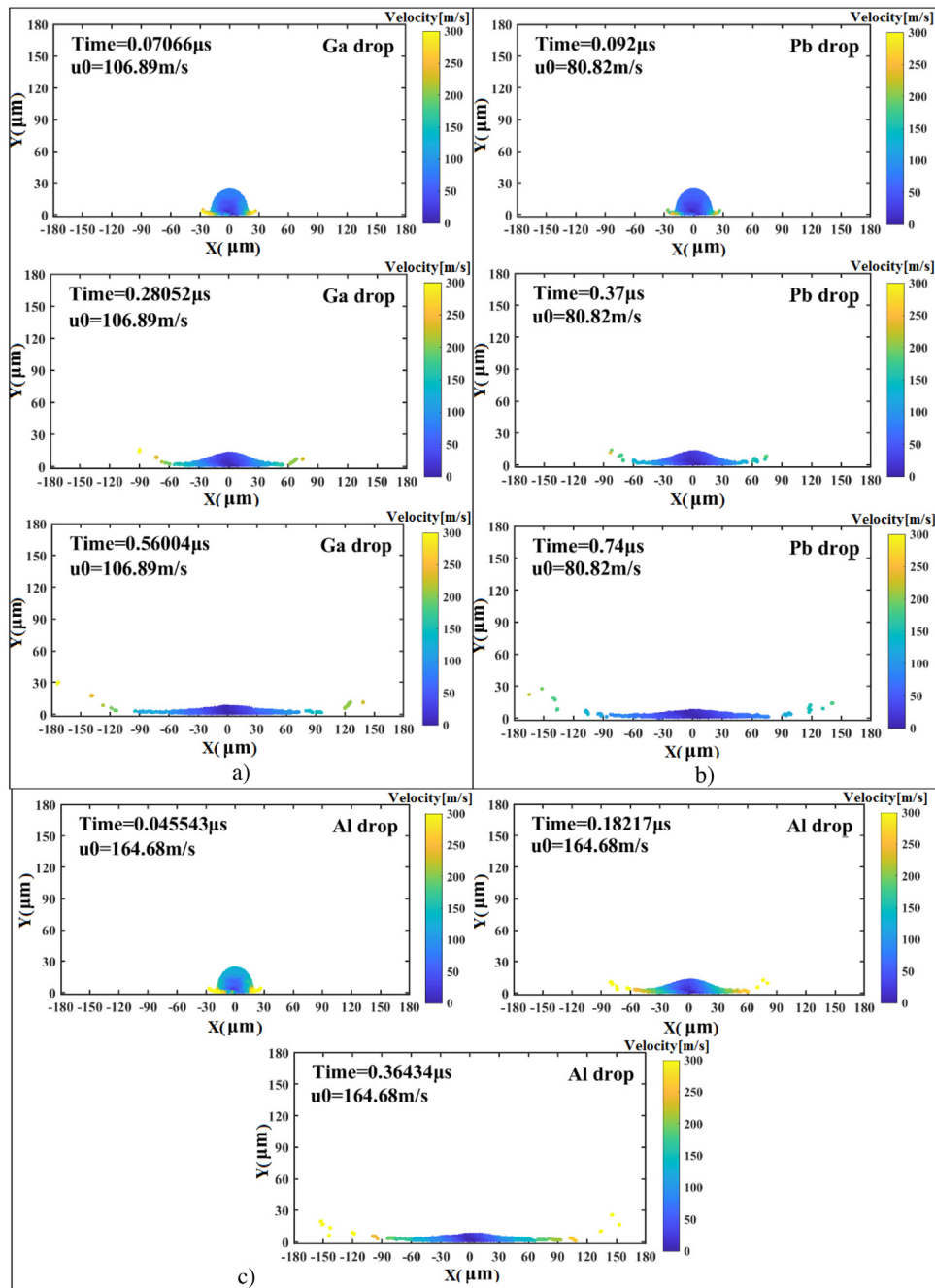


Fig. 8. Impact of 30 μm Ga, Pb and Al droplet with the same high incident kinetic energy ($\tau = 0.25, 1$ and 2).

Table 5
Parameters of droplet impingement with varied K .

Case	Material	d_0 (μm)	u_0 (m/s)	Re	We	Oh	E_k (μJ)	K
1	Sn	500	1	1818.2	6.6	0.00142	0.22809	142.3
2	Sn	500	3	5454.5	59.6	0.00142	2.05283	768.1
3	Sn	500	20	36363.6	2650.2	0.00142	91.23709	14117.9
4	Sn	2	150	1090.9	596.3	0.02238	3.28454E-4	1027.4
5	Sn	2	600	4363.6	9540.7	0.02238	0.00526	8622.9

factor is mainly determined by the viscosity while the spreading rate is affected by the surface tension. With the increase of viscosity, the maximum spread factor would decrease. And with the increase of surface tension, the spreading speed would slow down. When the incident kinetic energy is relatively high, the drop would splash.

Both the mass and kinetic energy ratio of secondary drop would increase with the increase of incident kinetic energy. Higher viscosity will also promote the splash of secondary drop.

2. A splash criteria of heavy droplet impingement is proposed by summarizing the numerical and experimental results.

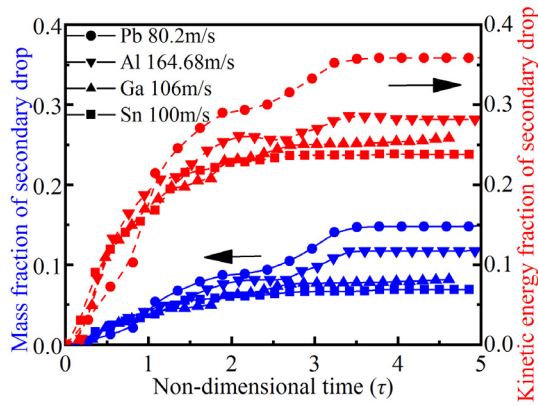


Fig. 9. Influence of droplet material on mass and kinetic energy fraction of secondary droplets (the shapes of symbol represent the droplet material; the red lines are the kinetic energy fraction of secondary drop and the blue lines are the mass fraction of secondary drop).

The criteria is given by combining the incident droplet Re number and Oh number: $K = Oh \cdot Re^{1.5346}$. The drop splashes when K reaches 3341.95, otherwise no splash occurs. This proposed splash threshold is able to estimate the outcomes of heavy droplet impinging on dry walls with different wall temperature (from room temperature to the boiling point).

3. The differences of splashing thresholds between heavy droplet and light droplet are discussed. The incident heavy drop is more difficult to splash than light drop within range of $Re < 4000$. When $Re > 4000$, obvious splash threshold still exists for heavy droplet impact, while the boundary becomes unclear for light droplet, making it hard to predict the outcome of drop impingement.

The current conclusions are obtained based on numerical results and the existing experimental data. The newly developed splash criteria of heavy droplet impingent provides a better understanding of heavy droplet dynamic behavior. The difference characteristics between light droplet and heavy droplet indicate that there is more to be investigated on the features of heavy

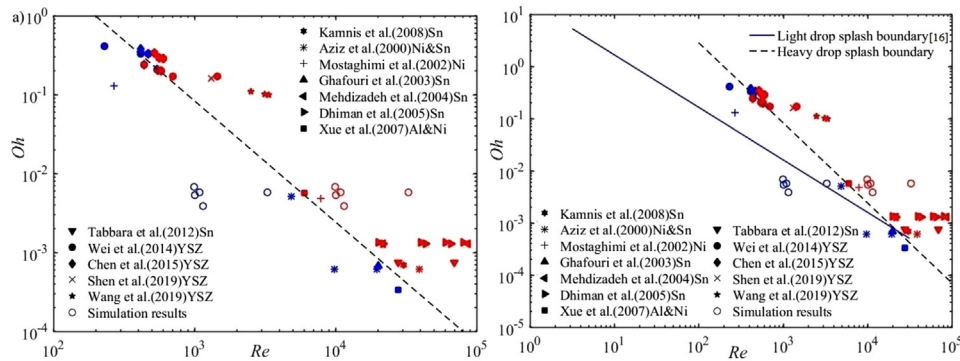


Fig. 10. Outcome and Splash criteria of heavy droplet/wall impingement (The shapes represent the information sources as shown in legend while the colors represent splash (red)/non-splash (blue) while).

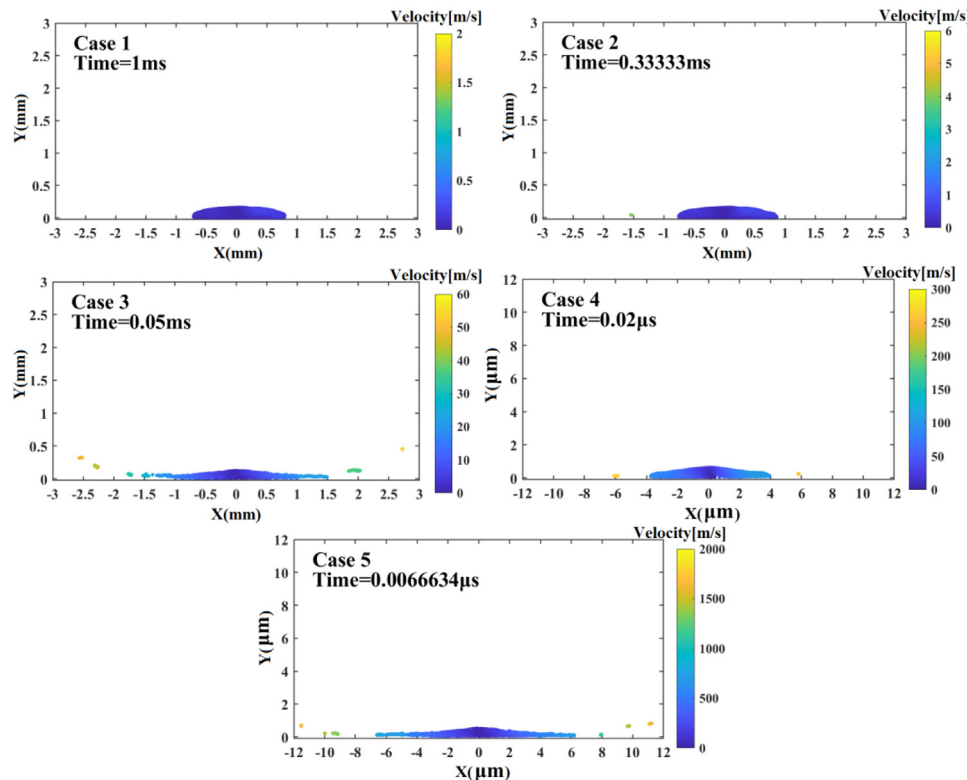


Fig. 11. Impact of Sn drops with different incident size and velocity ($\tau = 2$).

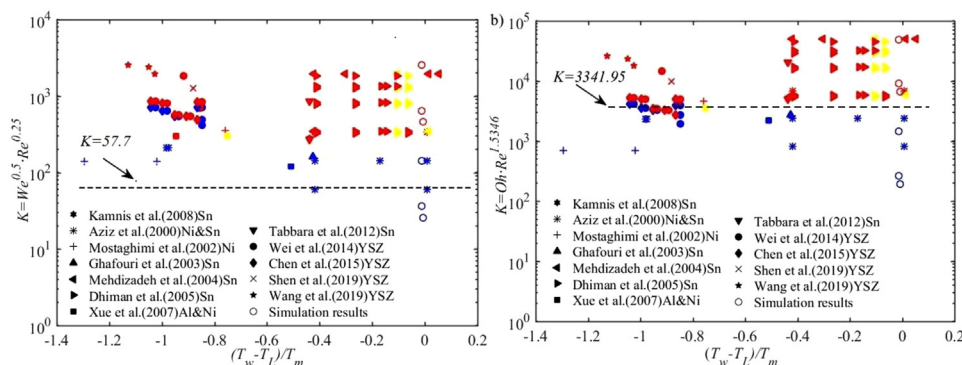


Fig. 12. Droplet behavior and splash threshold of heavy droplet impinging on dry wall with different wall temperature (the shapes represent the information sources as shown in legend while the colors represent splash (red)/suppressed splash (yellow)/non-splash (blue)).

droplet impactation. This result would help the experimental design of heavy droplet impinging on walls as well as the splash control of plasma spray, 3D printing, ejecta mixing, etc. However, more experimental data of metal drop impingement under different conditions is needed for further research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The work is completed with the support of National Natural Science Foundation Youth Fund of China (11702028), General Program (11772065) and the stable supporting funds of the Key Laboratory of Computational Physics, China.

References

- [1] G. Liang, I. Mudawar, Review of mass and momentum interactions during drop impact on a liquid film, *Int. J. Heat Mass Transfer* 101 (2016) 577–599.
- [2] C. Josserand, S.T. Thoroddsen, Drop impact on a solid surface, *Annu. Rev. Fluid Mech.* 48 (2016) 365–391.
- [3] A.L. Yarin, Drop impact dynamics: splashing, spreading, receding, bouncing..., *Annu. Rev. Fluid Mech.* 38 (2006) 159–192.
- [4] A.M. Worthington, R.S. Cole, *A Study of Splashes*, Longmans Green and Company, New York, 1908.
- [5] L.H.J. Wachters, N.A.J. Westerling, Boiling from liquid drops impact on a heated wall in the spheroidal state, *Chem. Eng. Sci.* 21 (11) (1966) 1047–1056.
- [6] J.D. Naber, R.D. Reitz, Modeling engine spray/wall impingement, *SAE Trans.* 97 (1989) 118–140.
- [7] C. Mundo, M. Sommerfeld, C. Tropea, Droplet-wall collisions: Experimental studies of the deformation and breakup process, *Int. J. Multiph. Flow.* 21 (2) (1995) 151–173.
- [8] C. Mundo, M. Sommerfeld, C. Tropea, On the modeling of liquid sprays impinging on surfaces, *Atomization Sprays* 8 (6) (1998) 625–652.
- [9] P. Schmidt, G. Knauss, Prallzerstä von flüssigkeiten bei nichtbenetzung, *Chem. Ing. Tech.* 48 (7) (1976) 659.
- [10] J. Senda, T. Kanda, M. Al-Roub, et al., Modeling spray impingement considering fuel film formation on the wall, *SAE Trans.* (1997) 98–112.
- [11] J. Senda, M. Ohnishi, T. Takahashi, et al., Measurement and modeling on wall wetted fuel film profile and mixture preparation in intake port of SI engine, *SAE Trans.* (1999) 1083–1098.
- [12] P.J. O'Rourke, A.A. Amsden, A spray/wall interaction submodel for the KIVA-3 wall film model, *SAE Trans.* (2000) 281–298.
- [13] R.L. Vander Wal, G.M. Berger, S.D. Mozes, The splash/non-splash boundary upon a dry surface and thin fluid film, *Exp. Fluids* 40 (1) (2006) 53–59.
- [14] J. Palacios, J. Hernández, P. Gómez, et al., Experimental study of splashing patterns and the splashing/deposition threshold in drop impacts onto dry smooth solid surfaces, *Exp. Therm Fluid Sci.* 44 (2013) 571–582.
- [15] H.J.J. Staat, T. Tran, B. Geerdink, et al., Phase diagram for droplet impact on superheated surfaces, *J. Fluid Mech.* (779) (2015).
- [16] T. Ma, L. Feng, H. Wang, et al., A numerical study of spray/wall impingement based on droplet impact phenomenon, *Int. J. Heat Mass Transfer* 112 (2017) 401–412.
- [17] S.D. Aziz, S. Chandra, Impact, recoil and splashing of molten metal droplets, *Int. J. Heat Mass Transfer* 43 (16) (2000) 2841–2857.
- [18] H. Tabbara, S. Gu, Modeling of impingement phenomena for molten metallic droplets with low to high velocities, *Int. J. Heat Mass Transfer* 55 (7–8) (2012) 2081–2086.
- [19] N.Z. Mehdizadeh, M. Raessi, S. Chandra, et al., Effect of substrate temperature on splashing of molten tin droplets, *J. Heat Transfer* 126 (3) (2004) 445–452.
- [20] R. Dhiman, S. Chandra, Effect of substrate temperature on splashing of molten metal droplets, in: *ASME 2004 Heat Transfer/Fluids Engineering Summer Conference*, American Society of Mechanical Engineers Digital Collection, 2004, pp. 71–78.
- [21] R. Dhiman, S. Chandra, Freezing-induced splashing during impact of molten metal droplets with high Weber numbers, *Int. J. Heat Mass Transfer* 48 (25–26) (2005) 5625–5638.
- [22] P. Wei, Z. Wei, S. Li, et al., Splat formation during plasma spraying for 8 mol% yttria-stabilized zirconia droplets impacting on stainless steel substrate, *Appl. Surf. Sci.* 321 (2014) 538–547.
- [23] D. Chen, Y. Wang, Y. Bai, et al., Influence of Reynolds number on droplet flattening behavior in plasma spraying, *J. Inorg. Mater.* 30 (01) (2015) 65–70.
- [24] Y. Wang, Y. Bai, K. Wu, et al., Flattening and solidification behavior of in-flight droplets in plasma spraying and micro/macro-bonding mechanisms, *J. Alloys Compd.* 784 (2019) 834–846.
- [25] X. Xu, J. Ouyang, T. Jiang, et al., Numerical analysis of the impact of two droplets with a liquid film using an incompressible SPH method, *J. Eng. Math.* 85 (1) (2014) 35–53.
- [26] X. Yang, S.C. Kong, 3D simulation of drop impact on dry surface using SPH method, *Int. J. Comput. Methods* 15 (03) (2018) 1850011.
- [27] V. Abdollahi, W.G. Habashi, M. Fossati, Multi-phase smoothed particle hydrodynamics modeling of supercooled large droplet dynamics for in-flight icing conditions, *Aerosp. Sci. Technol.* 82 (2018) 252–261.
- [28] X. Dong, X. Huang, J. Liu, Modeling and simulation of droplet impact on elastic beams based on SPH, *Eur. J. Mech. A Solids* 75 (2019) 237–257.
- [29] G. Hao, X. Dong, Z. Li, Numerical investigation of water droplet impact on horizontal beams, *Int. J. Mod. Phys. C (IJMPC)* 31 (08) (2020) 1–17.
- [30] R. Zhang, B. Zhang, Q. Lv, et al., Effects of droplet shape on impact force of low-speed droplets colliding with solid surface, *Exp. Fluids* 60 (4) (2019) 64.
- [31] G.A. Mesyats, N.M. Zubarev, Hydrodynamics of the molten metal in a vacuum arc cathode spot at near-threshold currents, *J. Appl. Phys.* 113 (20) (2013) 203301.
- [32] W.T. Buttler, D.M. Oró, R.T. Olson, et al., Second shock ejecta measurements with an explosively driven two-shockwave drive, *J. Appl. Phys.* 116 (10) (2014) 103519.
- [33] J.J. Monaghan, SPH without a tensile instability, *J. Comput. Phys.* 159 (2) (2000) 290–311.
- [34] J.P. Gray, J.J. Monaghan, R.P. Swift, SPH elastic dynamics, *Comput. Methods Appl. Mech. Engrg.* 190 (49–50) (2001) 6641–6662.
- [35] J.U. Brackbill, D.B. Kothe, C. Zemach, A continuum method for modeling surface tension, *J. Comput. Phys.* 100 (2) (1992) 335–354.
- [36] H.F. Qiang, F.Z. Chen, W.R. Gao, Modified algorithm for surface tension with smoothed particle hydrodynamics and its applications, *Comput. Model. Eng. Sci.* 77 (3) (2011) 239.

- [37] Y.Q. Han, H.F. Qiang, J.L. Zhao, et al., A new repulsive force model for solid wall treatment with smooth particle hydrodynamics, *J. Phys.* 62 (04) (2013) 334–344.
- [38] G.R. Liu, M.B. Liu, *Smoothed Particle Hydrodynamics: A Meshfree Particle Method*, World scientific, 2003.
- [39] J.P. Morris, Simulating surface tension with smoothed particle hydrodynamics, *Internat. J. Numer. Methods Fluids* 33 (3) (2000) 333–353.
- [40] S. Shakeri, S. Chandra, Splashing of molten tin droplets on a rough steel surface, *Int. J. Heat Mass Transfer* 45 (23) (2002) 4561–4575.
- [41] R.I.L. Guthrie, T. Iida, *The Physical Properties of Liquid Metals*, Clarendon press, 1987.
- [42] J. Mostaghimi, M. Pasandideh-Fard, S. Chandra, Dynamics of splat formation in plasma spray coating process, *Plasma Chem. Plasma Process.* 22 (1) (2002) 59–84.
- [43] R. Ghafouri-Azar, S. Shakeri, S. Chandra, et al., Interactions between molten metal droplets impinging on a solid surface, *Int. J. Heat Mass Transfer* 46 (8) (2003) 1395–1407.
- [44] M. Xue, Y. Heichal, S. Chandra, et al., Modeling the impact of a molten metal droplet on a solid surface using variable interfacial thermal contact resistance, *J. Mater. Sci.* 42 (1) (2007) 9–18.
- [45] M. Shen, B.Q. Li, Q. Yang, et al., A modified phase-field three-dimensional model for droplet impact with solidification, *Int. J. Multiph. Flow.* 116 (2019) 51–66.
- [46] M. Qin, C. Tang, S. Tong, et al., On the role of liquid viscosity in affecting droplet spreading on a smooth solid surface, *Int. J. Multiph. Flow.* 117 (2019) 53–63.
- [47] G.T. Liang, Y.L. Guo, S.Q. Shen, Observation and analysis of wetting spherical surface caused by low velocity impact of droplet, *J. Phys.* 62 (18) (2013) 328–333.